

Pedagogical Frameworks for 3D Gaussian Splatting in University Education: Navigating Emerging Technologies in Cross-Disciplinary Learning Environments

Lukasz Mirocha
School of Creative Media
City University
Kowloon, Kowloon, Hong Kong
Opus Lab
Hong Kong, Hong Kong
mirocha.lukasz@gmail.com

Abstract

This talk examines the integration of 3D Gaussian Splatting (3DGS) technology into creative media, design and journalism education programs at undergraduate and postgraduate levels. It presents findings and best practices from multiple educational implementations across different institutional contexts and approaches. As an emerging spatial reconstruction technique that has rapidly gained attention in computer graphics research and creative industries, 3DGS presents unique opportunities and challenges for educators seeking to incorporate cutting-edge technologies into curricula. This talk draws from direct experience implementing 3DGS education across various formats and cultural contexts, including intensive workshop-based programs, semester-long courses, and professional development sessions, conducted at institutions and conferences worldwide, including Hong Kong, Poland, Japan, Korea, and the United Arab Emirates.

CCS Concepts

• Collaborative learning; • Media arts; • Software evolution;

Keywords

3D Gaussian Splatting, Media Design Education, Design Pipelines, Curriculum Design

ACM Reference Format:

Lukasz Mirocha. 2025. Pedagogical Frameworks for 3D Gaussian Splatting in University Education: Navigating Emerging Technologies in Cross-Disciplinary Learning Environments. In *SIGGRAPH Asia 2025 Educator's Forum (SA Educator's Forum '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757372.3771867>

1 Introduction to 3D Gaussian Splatting in Educational Context

3D Gaussian Splatting represents a significant advancement in neural rendering techniques, building upon the foundational work of

Kerbl et al. [Kerbl, 2023] The technique enables rather straightforward 3D reconstruction of objects and locations based on an aligned set of photos or video frames. The technology's accessibility and relatively modest computational requirements have facilitated its rapid adoption across various creative disciplines. The educational relevance of 3DGS extends beyond its technical merits, with recent applications in cultural heritage preservation [Dahaghin, 2024], and immersive media creation [Haslbauer et al., 2025], [Qiu, 2025] demonstrating the technology's potential to bridge traditional creative practices with emerging digital methodologies. Educational institutions should recognize 3DGS as a component of contemporary media literacy, particularly in programs focused on future-oriented practices, leveraging reality capture, digital twins, volumetric video, XR/3D and interactive content while fostering critical engagement with AI-driven tools.

2 Educator Profile and 3DGS Educational Activities

Since 2024, I have taught 3DGS across Asia, Europe, and North America, introducing diverse audiences to its theory and practice. Using pipelines from smartphone scanning apps to advanced CUDA-based software [Niantic et al., 2024], I engaged students from design, journalism, and creative media. This include masterclasses at SWPS University for design students, a journalism course at University of National Education Commission, and a semester-long City University of Hong Kong course conducted with industry partners collaboration (2024 & 2025). I also led masterclasses for professionals at SIGGRAPH Asia 2024 and ISEA 2025 as well as several in-house masterclass sessions for the game development and VFX industries.

3 Challenges in Emerging Technology Education

The rapid pace of development in 3DGS tools creates a dynamic landscape. Unlike established creative software with comprehensive documentation, stable feature sets, and certifications, 3DGS tools often exist in experimental states with limited documentation and frequent updates that disrupt established workflows. The non-standardized nature necessitates collaborative project development. Students possess diverse skill sets from cinematography and video editing to 3D modeling and programming. This diversity becomes

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SA Educator's Forum '25, Hong Kong, Hong Kong
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2132-8/2025/12
<https://doi.org/10.1145/3757372.3771867>

an asset when channelled through group-based structures requiring contributions from storytellers, camera operators, editors, 3D artists, and technical specialists comfortable with command-line interfaces and experimental repositories. Educational implementations at SWPS University and University of National Education Commission were conducted in intensive one-week formats, requiring accelerated learning. These "hackathon-style" implementations demanded careful scaffolding and substantial hands-on guidance to enable meaningful outcomes within compressed timeframes – open-themed video shorts integrating 3DGS sequences. Conversely, the semester-long City University of Hong Kong implementation allowed gradual skill development, with students allocated one month (2024) or two months (2025) for 3DGS project completion. This enabled combining 3DGS with 3D tools, game engines and post-production tools for first-hand experience of emerging design pipelines.

4 Pedagogical Approaches and Methodological Considerations

Two primary pedagogical approaches emerged: standalone masterclass formats and integrated curriculum components. Standalone masterclasses focus intensively on 3DGS-specific skills, including capture methods, processing workflows, and integration techniques for incorporating outputs into traditional video production pipelines. These sessions emphasize hands-on technical competency development and provide concentrated exposure to applications and case studies. This format enables technical engagement, making students capable of immediate practical application at foundational levels (e.g. using smartphone-based apps). Integrated curriculum approaches position 3DGS as one component within broader explorations of emerging media technologies. In the semester-long course, 3DGS instruction occurs alongside generative AI video techniques, game engine real-time rendering, and other contemporary production technologies. This approach facilitates comparative analysis and enables students to develop informed perspectives on relative strengths and appropriate applications of different emerging technologies. It fosters critical thinking about technology adoption and prepares students to navigate rapidly evolving creative technologies through self-learning. The selection between approaches depends on institutional context, student populations, and learning objectives. Assessment methods require consideration of technical competency and creative application, while leaving necessary leeway for student experimentation with non-established medium. Student project emphasize process documentation alongside final outputs, recognizing that experimental technologies often produce unexpected results that may not conform to traditional creative project expectations. Group-based, course leader and peer-assessment exercises acknowledge the collaborative nature of 3DGS workflows while ensuring individual contribution recognition.

5 Conclusion

The integration of 3DGS into diverse education programs represents both opportunity and challenge. The technology's constant development and widening applications create compelling reasons for curricular inclusion, while its experimental nature demands

flexible pedagogical approaches. Successful implementations require attention to student skill diversity, collaborative project structures for peer-learning, and adaptive assessment. The talk shares best practices for introducing emerging technology into university curriculum and invites discussion with fellow educators on their challenges with adopting digital technologies into creative media / design / humanities curricula. Sample student works will be presented.

References

- Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Trans. Graph.* 42, 4, Article 139 (August 2023), 14 pages. <https://doi.org/10.1145/3592433>.
- Dahaghin, Mahtab, Myrna Castillo, Kourosh Riahidehkordi, Matteo Toso, and Alessio Del Bue. 2024. Gaussian Heritage: 3D Digitization of Cultural Heritage with Integrated Object Segmentation. *arXiv preprint arXiv:2409.19039*.
- Haslbauer, Philip *et al.* 2025. VR Planica: Gaussian Splatting Workflows for Immersive Storytelling. In *Proceedings of the ACM International Conference on Interactive Media Experiences Workshops*, June 03, 2025, Niterói/RJ, Brasil. SBC, Porto Alegre, Brasil, 47-53. DOI: <https://doi.org/10.5753/imxw.2025.8141>.
- Qiu, Shanshan, Bingchen Xie, Qiuyu Liu, and Pheng-Ann Heng. 2025. Advancing Extended Reality with 3D Gaussian Splatting: Innovations and Prospects. In *Proceedings of the 2025 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR '25)*, Lisbon, Portugal. IEEE, 203–208. <https://doi.org/10.1109/AIxVR63409.2025.00039>.
- Niantic. 2024. Scaniverse.; LumaLabs. 2024. Luma 3D; Jawset. 2024. Postshot. nerfstudio Team. 2024. Nerfstudio.